

IoT-Enabled Aquaculture System with Feeding, Mortality Detection, Budgeting, Monitoring, and Alerts

Project ID: 25-26J-295

Project Proposal Report

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BSc (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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August 2024

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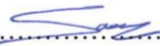
DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This feature gives a Smart Maintenance Reminder and Water Quality Monitoring System that uses sensors to change the schedule for pond chores instead of having them set on a calendar. A compact, waterproof edge controller (ESP 32) constantly collects turbidity (SEN0189), pH (Atlas Scientific), temperature, and optional dissolved oxygen signals and sends them to a mobile or web app. A lightweight analytics layer uses threshold checks, trend analysis, and rate-of-change analysis to find drifts and sudden changes, like when turbidity spikes after heavy feeding or pH levels change after it rains. When parameters move outside of safe ranges or show early signs of deterioration, the system updates the farmer's digital task calendar, sends usage-based reminders (for things like changing filters or replacing oxygen stones), and sends short alerts. The system reminds the user when treatment is needed and keeps a complete record of every action (timestamp, dosage, notes/photo) to keep an audit trail, even though chemical dosing is not automated. It also saves completed tasks and sensor histories for benchmarking and ongoing improvement. The system cuts down on missed maintenance, saves resources, and helps fish stay healthy, especially for small businesses that only have one pond to manage. It does this by combining real-time monitoring with reminders that are useful and aware of the situation.

Keywords: monitoring water quality, turbidity, dissolved oxygen, adaptive maintenance, scheduling tasks

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LIST OF ABBREVIATIONS

Abbreviation	Description
IoT	Internet of Things
CBM	Condition Based Maintenance
DO	Dissolved Oxygen
ESP 32	Espressif 32-bit Microcontroller
pH	Potential of Hydrogen
CBM	Condition-Based Maintenance
WQ	Water Quality
TLS	Transport Layer Security
FCM	Firebase Cloud Messaging

1 INTRODUCTION

1.1 Background study

In order to maintain fish healthy and productive, aquaculture ponds require to have very strict control over the quality of the water. In real life, farmers must keep track of a lot of things at once, like turbidity (clarity), pH,[1] temperature, and often dissolved oxygen (DO). They also must do things such as clean filters, replace oxygen stones, remove sludge, and do chemical treatments every so often. Most of the time, traditional maintenance is based on a schedule (for example, "clean filters every two weeks"), which doesn't consider how quickly things can change after things like heavy feeding, algae blooms, sudden rain, or warm spells. Because of this, tasks are sometimes done too late (after quality has already gone down) or too early (wasting time and materials), and small problems can turn into fish stress or disease if they aren't noticed between manual checks.

Low-cost IoT hardware that has come out recently makes it possible for smallholders to keep an eye on their ponds all the time. A small, waterproof edge controller, like a ESP 32, can read turbidity (SEN0189), pH (Atlas Scientific), temperature [2], and optional DO in almost real time and send these values to a phone or web dashboard. Continuous streams do more than just set off simple threshold alarms. The system can see early signs of deterioration, like a turbidity spike after heavy feeding or a slow drift in pH towards dangerous levels, before the fish are affected. Usage-aware logic links maintenance to the state of the pond instead of the calendar, so reminders come when they are really needed. A digital record of completed tasks and dosing logs [3] (including time, amount, reason, and an optional photo) creates a reliable audit trail that can be used to learn and trace things across cycles and seasons.

This feature makes that method official by calling it a Smart Maintenance Reminder and Water-Quality Monitoring System [6]. We constantly collect sensor data and compare it to safe bands. A lightweight analytics layer checks for trends, persistence, and anomalies. When conditions change or go beyond certain limits, [7] the virtual assistant updates the farmer's task calendar, sends usage-based notifications (like "replace filter media" or "inspect aeration"),[8] and suggests chemical treatment when it's needed (note: dosing is still a manual decision made in the app; the system does not auto-dose). Every action and reading are saved so that you can compare them and keep getting better. The module connects to other project functions, like feeding automation (to explain turbidity changes after feeding) and budget tracking (to attach costs to tasks), so that tasks, costs, and sensor insights stay in sync. The system helps small pond operations avoid forgetting to do chores, save resources, and keep the water conditions healthier with minimal effort by replacing fixed intervals with sensor-driven, [9]adaptive maintenance.

1.2 Literature survey

IOT water quality monitoring in aquaculture.

Low-cost IoT nodes (ESP 32) with pH, temperature, DO, and more recently turbidity sensors make it possible for small farms to keep an eye on their ponds all the time. Research emphasizes basic edge conditioning (filtering, debouncing, temperature compensation) and regular sensor maintenance [1] (pH calibration, DO membrane/electrolyte, turbidity fouling) to ensure data reliability.

Beyond static thresholds: trend and persistence

Older systems only had hard limits. Newer systems check for trends, slopes, and persistence to find problems sooner and stop "chatter" from happening. Moving averages, rolling z-scores, hysteresis, and change-point detection can help tell the

difference between short bursts of noise and real changes in the regime (for example, turbidity spikes after feeding or gradual pH drift).

Condition-based maintenance (CBM)

According to maintenance literature, calendars waste time and money or do things too late. CBM connects tasks to measured conditions, like checking filters when turbidity stays high or checking aeration when DO levels drop repeatedly. This turns raw signals into clear, scheduled maintenance triggers instead of random alarms.

Dosing and auditing with a person in the loop

Auto-dosing in open ponds is dangerous and could lead to lawsuits, so the best way to do it is to have the system recommend treatment, the farmer approve it, and then do it. Keeping track of each action (time, dose, reason, photo) makes a record of dosing and maintenance that can be used to find the root cause and make things better all the time.

integration and deployment pattern

Feeding events, rainfall, and aeration regimes all affect water quality signals. To make the most of this, it's a good idea to share context between modules [2] (for example, feeding → turbidity tolerance; rain → stricter pH watch; budget → reset consumable counters). Edge analytics for quick alerts that work even when the internet is down and cloud backends for dashboards, history, and access for multiple users are two common patterns.

The gap that drives this function

Many aquaculture tools only have real-time charts, and many maintenance apps don't have sensor integration. Auto-dosing is also rare. There is a need for a cheap, sensor-based system that can analyse turbidity, pH, temperature, and optional DO [3] in real time, turn trends into task updates and replacement reminders, keep a full dosing and maintenance log, and share events with feeding and budgeting modules. This would make condition-based, adaptive maintenance possible for small ponds.

1.3 Research gap

Most aquaculture IoT systems focus on constant sensing and alarms for pH, temperature, and (less often) DO or turbidity levels. They show live charts and send push alerts, but they don't often turn sensor trends into a condition-based maintenance (CBM) plan. This means that when the data says the pond needs it, they automatically update a task list with things like cleaning the filter, replacing the oxygen stone, and removing sludge. There are separate maintenance apps (CMMS), but they are based on calendars and don't have pond telemetry. Also, the farmer usually does the dosing without a structured log of the dosing and maintenance that is linked to the sensor history. This makes it harder to trace and learn. Lastly, turbidity is often not measured enough, even though it is a quick and easy way to measure post-feeding residue and suspended solids [4].

The gap is a low-cost, all-in-one assistant that, keeps an eye on turbidity, pH, temperature, and optional DO in real time; uses trend/persistence logic (not just hard limits) to find problems early; sends CBM [10]reminders by automatically updating a digital task calendar; keeps a human-in-the-loop dosing workflow with a full audit trail (timestamp, dose, reason, photo); and shares events with feeding and budgeting modules so that tasks, costs, and sensor insights stay together. Function 4 is meant to fill this gap.

Yes = ✓

No = ✗

Partial/unclear = ~

Study/year	setting	Sensors covered(pH/Temp/DO/Turbidity)	Trend & persistence (beyond thresholds)	CBM reminders/task scheduling	Dosing log & audit trail	Key limitation & new function
Saha et al./20218(IOT fish farm monitoring)	Aquaculture Ponds	pH Temp DO	~	✗	✗	No turbidity channel, no sensor-driven maintenance or dosing record; just live monitoring and alerts.
Shete et al.,2024(IOT WQ monitoring & prediction)	Aquaculture	pH Temp DO	~	✗	✗	It adds analytics and predictions, but not a CBM task engine. Dosing and maintenance are not logged or linked to trends.
Abdullah et al.,2024(Aquaculture IOT overview)	Review	pH Temp DO	~	✗	✗	Only surveys; no telemetry-based maintenance workflow or dosing audit.
Jardine et al.,2006(CBM review,industry)	General CM	~	✓	~	✗	Establishes CBM principles but not for ponds; does not include water quality sensing or turbidity context.
Auto-dosing controllers (industrial/lab)	Tanks/RAS	pH Temp DO	~	✗	✗	Stress automated dosing (safety/liability); no full audit workflow or CBM reminders.
Proposed system (this work)	Aquaculture Ponds	pH Temp DO Turbidity	✓ (trend, rate,	✓ (sensors-driven CBM)	✓	Integrates multi-sensor trends → adaptive task calendar, prompts for treatment, and keeps a complete audit trail; links

			persiste nce)			with feeding & budget modules.
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Table 1.1: Comparison chart

1.4 Research problem

Small pond operations still don't have a low-cost, all-in-one way to use continuous water-quality measurements to make maintenance decisions based on the conditions. Most of the IoT tools for aquaculture only have live charts and alarms that go off when certain thresholds are met. Generic maintenance apps, on the other hand, use calendar schedules that don't consider how turbidity, pH, temperature, and dissolved oxygen (DO) levels change after things like feeding or rain. This leads to chores that are late or not needed, a lack of dosing traceability, and water-quality drift that can be avoided and hurts fish health. The research problem is to create and test a sensor-driven virtual assistant that continuously checks turbidity, pH, temperature, and optional DO; uses trend, rate-of-change, and persistence logic (not just hard limits) to find early signs of deterioration; automatically updates a digital task calendar with usage-based reminders (e.g., filter cleaning, oxygen-stone replacement, sludge removal); and supports human-in-the-loop dosing with a full audit trail (timestamp, dose, reason, photo). The solution must work well in outdoor pond conditions, be able to ignore sensor noise and fouling with simple edge filtering and calibration, work even when the connection is spotty (edge-first alerts, cloud sync), and send short mobile notifications that farmers who aren't experts can act on quickly. It should also share events with the feeding and budget modules so that tasks, costs, and sensor data stay in sync.

2 RESEARCH OBJECTIVE

2.1 Main objective

To develop and validate a low-cost, sensor-driven virtual assistant that turns continuous data about the quality of pond water (like turbidity, pH, temperature, and optional DO) into reminders for condition-based maintenance (CBM) and human-in-the-loop dosing prompts, with clear mobile alerts, a full dosing/maintenance log, and easy integration with feeding and budgeting modules for small outdoor ponds.-

2.2 Specific objective

- Implement more than one sensor to keep an eye on things. Set up a waterproof edge node to continuously collect turbidity (SEN0189), pH (Atlas Scientific), temperature, and optional DO, with basic signal conditioning (filtering, debouncing, and temperature compensation).
- Develop analytics for trends and persistence. Use rate-of-change, moving averages, and persistence windows to find early signs of deterioration that go beyond static thresholds. For example, a steady rise in turbidity after feeding or a slow drift in pH.
- Build a task engine for CBM. When trend rules are met, automatically update a digital task calendar (for example, for cleaning filters, replacing oxygen stones, or removing sludge) instead of at set times. Include logic for priority and due dates.
- Provide alerts and a good user experience. Send short push notifications with a maximum end-to-end latency of 15 seconds. Use hysteresis and cool-downs to stop duplicate "alert chatter."
- Support with sensor maintenance and calibration. Add quick calibration flows and reminders for sensor health based on how often they are used and how long

they have been used (for example, changing the DO membrane/electrolyte or cleaning the turbidity lens).

3 METHODOLOGY

Overview

We use a build-test-refine cycle to turn streams of continuous turbidity, pH, temperature, and optional DO into condition-based maintenance (CBM) reminders and prompts for human-in-the-loop dosing. A Raspberry Pi edge node that is waterproof collects sensor data, does light analytics (threshold, trend, rate of change, persistence), and sends alerts to the cloud or to the local network. A task engine turns detections into calendar items that have due dates and priorities. All readings, tasks, and dosing actions are kept for review and to make things better all the time.

Hardware Configuration

There is an IP-rated box near the pond that holds a ESP 32 (WROOM S3 or something similar). There are sensors for turbidity (SEN0189) using an ADC (like an ADS1115), pH (using an Atlas Scientific interface), temperature (like a DS18B20), and optional DO (using an Atlas Scientific interface). Cable glands and strain relief keep things from getting in, and analogue grounds are separate from pump power circuits. Wi-Fi is used for backhauling, and the node gets power from a fused DC supply or battery (with an optional solar trickle). There is a simple status LED and a physical reset on the enclosure.

Sensor calibration & care

Commissioning includes two-point (or multi-point) calibration for pH (buffer solutions), turbidity (formazin / standard or relative calibration with clear water + known cloudy sample), and DO (air-saturation and zero-oxygen solutions) where used. pH/DO is adjusted for temperature. The system keeps track of sensor health counters (days/uses since last calibration, membrane/electrolyte age, turbidity lens fouling risk) and automatically makes tasks for probe care.

Analytics: Threshold, Trend, and Persistence

There are safe bands and early-warning bands for each parameter. The edge computes beyond static limits:

Slope or rate of change (Δ over a rolling window) to see when things are getting worse, like when turbidity rises after feeding or pH slowly drifts.

Persistence (time spent outside the band) to avoid reacting to short spikes.

Windows of context For example, the turbidity watch could be less strict right after a feeding event from the feeding module, and the pH watch could be stricter after it rains if it is integrated.

Events are given a score (severity = function of deviation $\times$ persistence) and only confirmed when confidence is above a certain level.

Data management, safety, and privacy

First, the edge node writes sensor streams (turbidity, pH, temperature, and optional DO), task states, and dosing logs. Then, they are synced to the cloud using store-and-forward, which keeps the order and timestamps. TLS is used for all traffic, keys are

safe, and access is based on roles (owner/manager/viewer) with a log of edits and completions. Raw data is kept for a set amount of time (then down sampled), and dosing photos are compressed and have their location data removed. The system only gathers the most important account information, lets users export and delete their accounts on request, and never auto-doses; prompts need to be confirmed by a person.

Operational workflow

When the node boots up, it tests its sensors, loads calibration and thresholds, and connects to Wi-Fi. The loop reads the sensors, updates the smoothed values, checks the rules, and sets up tasks or alerts as needed. If you're not connected to the internet, alerts and data will be stored locally and synced automatically when you reconnect.

Evaluation Plan

We will set up pond-like tests to make sure that trend-based warnings go off before hard limits (like a steady rise in turbidity or a gradual drift in pH). These tests will happen at least one decision interval earlier than static thresholds. Persistence and hysteresis will be used to adjust false positives to less than one per day. Impact will be measured by a calendar baseline showing a 20% or more improvement in completing tasks on time and a clear recovery trend after the task is done. Reliability goals: at least 24 hours of continuous uptime with automatic recovery, an alert latency of 15 seconds or less, and no data loss during outages through store-and-forward. Targets for process quality: at least 95% of dosing actions logged (with a timestamp, dose, reason, and photo) and following calibration prompts with fewer sensor drift incidents.

3.1 System overview diagram

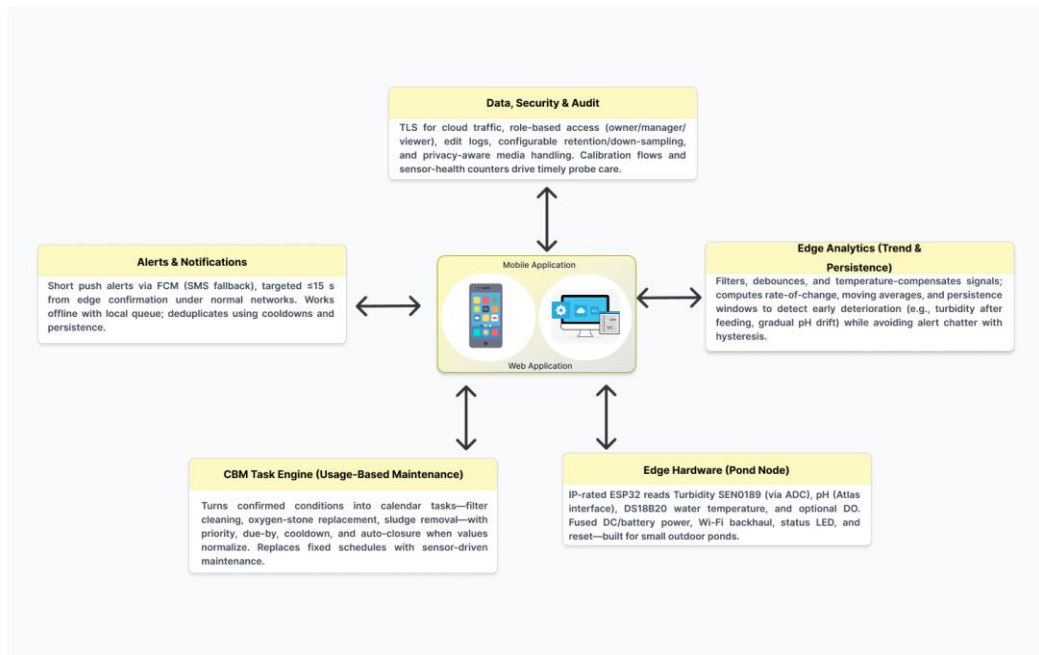


Figure 3.1: System overview

3.2 Component overview diagram

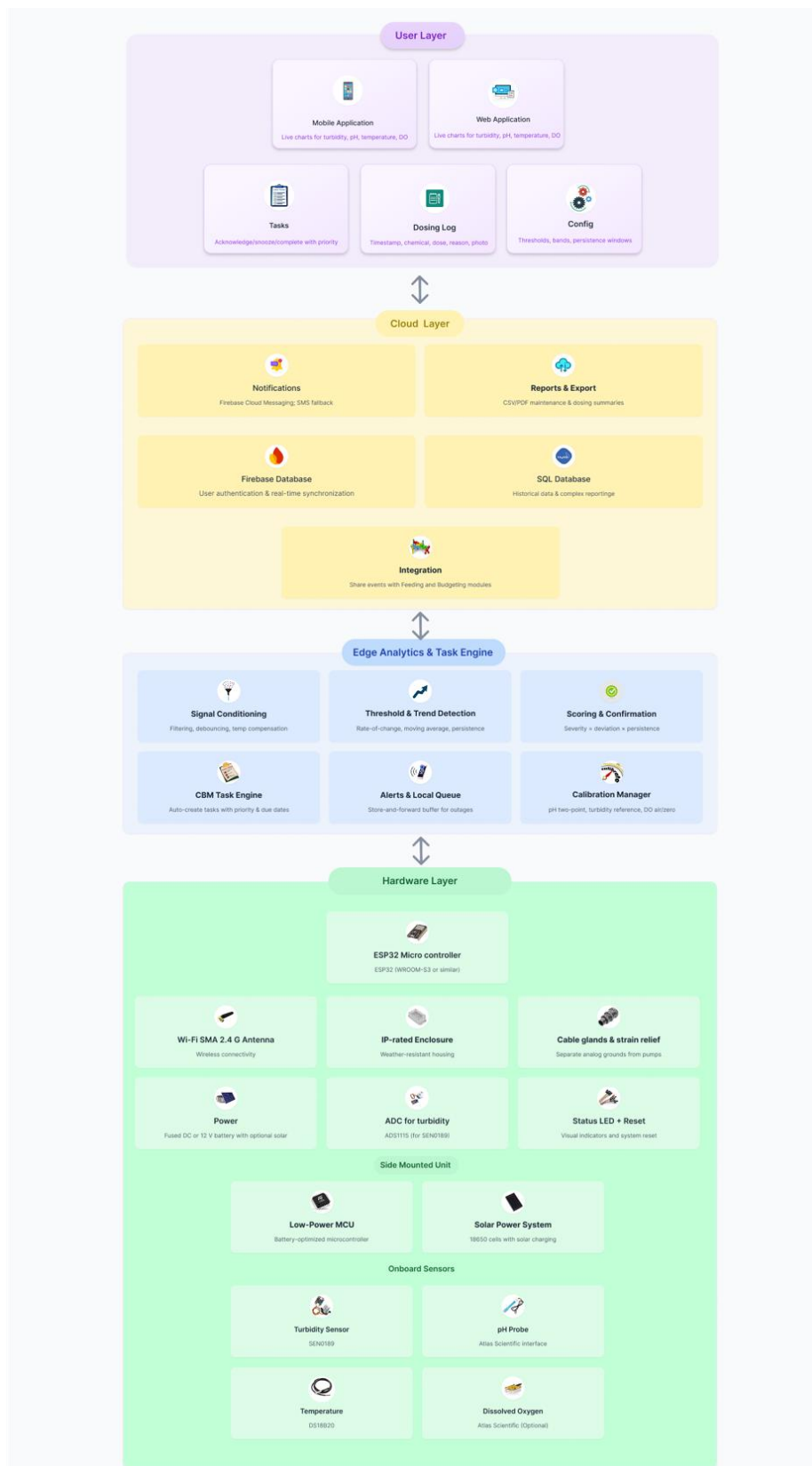


Figure 3.2: component overview

3.3 Gantt chart

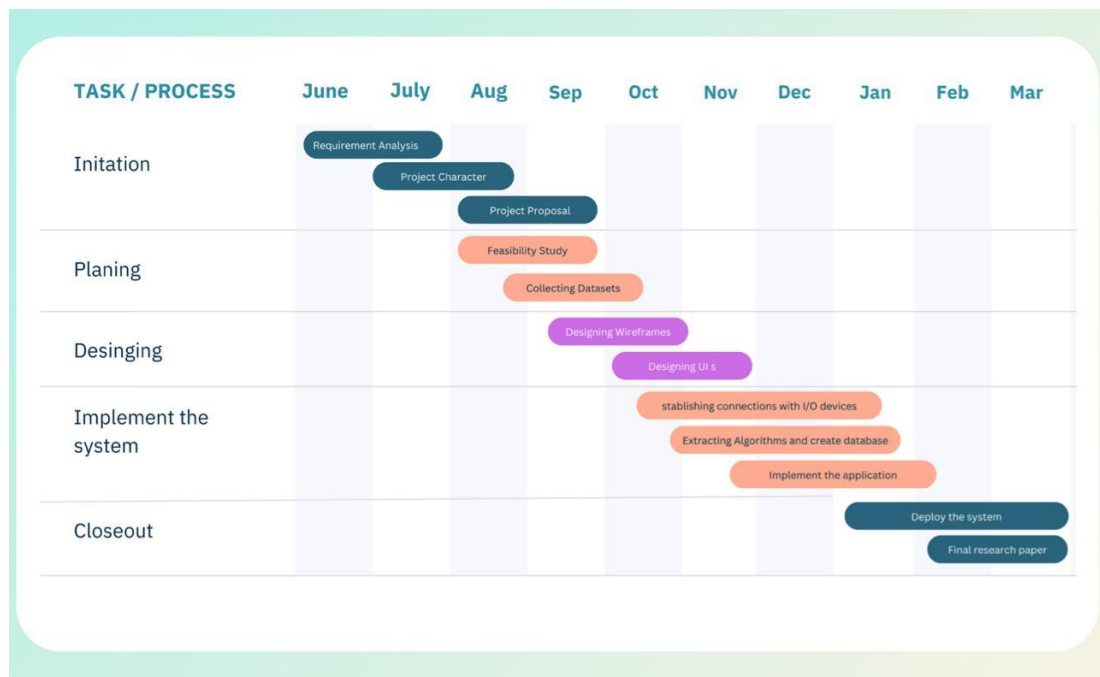


Figure 3.3: Gantt chart

4 PROJECT REQUIREMENTS

4.1 User requirements

- View turbidity, pH, temperature, and optional DO in real time on your phone.
- Get condition-based maintenance assignments, not merely threshold alerts.
- See the due-by priority and use a single tap to acknowledge, snooze, or finish tasks.
- Log time, dose, reason, and photo; receive dosing prompts but maintain human confirmation.
- View the reminders for sensor care (turbidity cleaning, DO membrane, pH calibration).
- Operate without losing data during outages by using local alerts and later syncing.

- Export dosing and maintenance reports and manage who can view and edit them.
- Modify thresholds and safe bands without using code.

4.2 Functional requirements

- Multi-sensor acquisition. Using the right interfaces and ADC, a Raspberry Pi edge node can read turbidity (SEN0189), pH (Atlas Scientific), temperature (DS18B20), and DO (optional).
- Calibration and compensation. Give quick flows for pH (buffers), turbidity (clear/known sample), DO (air/zero), and temperature compensation where needed. Store the factors.
- Pre-processing the edge. Add a timestamp, filter, and smooth signals (moving average/EWM), and use outlier rejection, debouncing, and hysteresis to stop alert chatter.
- Threshold and trend analysis. Check safe bands and trend/rate-of-change/persistence windows to find signs of early deterioration, like a spike in turbidity after feeding or a slow drift in pH.
- The CBM task engine. Set confirmed conditions into tasks, like cleaning the filter, replacing the oxygen stone, and removing the sludge. Give each task a priority, a due date, a cool-down period, and an automatic closure when the values return to normal.
- Calendar and reminders for tasks. Keep a digital calendar, send reminders to your phone or the web, and let people acknowledge, snooze, and finish tasks with notes.
- Notifications and alerts. Send short push alerts (parameter, direction, magnitude, duration, suggested action, task link) through FCM. You can also send SMS messages as a backup. Group or avoid duplicates.

- Managing configurations. Safe bands that can be changed, early-warning bands, trend/persistence windows, context windows, recipients, and alert cool-downs; they stay the same after restarts.
- Access control. Roles (owner, manager, viewer); a basic audit of who did what or logged their doses.
- Exporting and reporting. Make PDF and Excel files of maintenance and dosing summaries and trend snapshots for review and comparison.

4.3 Non - functional requirements

- Detection timeliness: Early-warning alerts should go off before hard limits, and the time it takes for a mobile alert to go off from start to finish should be less than 15 seconds (typical network).
- Accuracy and stability: in the expected probe specs between calibrations.
- Availability and strength: Continuous operation for at least 24 hours without human intervention; watchdog auto-recovery; no data loss with store-and-forward.
- Footprint of performance: Works on ESP 32 (no GPU); average CPU/RAM usage is less than or equal to 70% when the system is stable.
- Usability: Creating a task is automatic, and it takes less than 15 seconds to complete a task or log into a dose on the phone. Alerts are short and useful.
- Safety and privacy: TLS for cloud traffic; safe storage of keys; access based on roles; configurable retention for sensor and dosing records.
- Maintainability: Modular code for tasks, alerts, analytics, and acquisition; rules that are driven by configuration; and clear logs for fixing problems.

4.4 Test cases

Test Case ID	Task	Test Objective	Expected Result	Status
TC_01	Turbidity Increase After Feeding	Detect post-feeding turbidity spike and raise maintenance task.	If turbidity stays above X NTU for $\geq Y$ minutes after feeding, a high-priority "Filter inspection/cleaning" task is created within 5 minutes. No task if $< Y$.	Passed
TC_02	Early Warning for pH Drift	Warn about a slow pH decline before it breaches safe limit.	Alert: "early pH drift" and task: "check buffer/partial water change" (medium). Trigger happens ≥ 30 mins before hard limit with hysteresis.	Passed
TC_03	Night-time DO Dips Detection	Raise alert and review task if DO falls during night period.	If DO $<$ threshold for Z mins at night, alert with tag "night-time" and task "review aeration schedule" triggered within 15s. No alert if DO recovers quickly.	Passed
TC_04	Detecting Jam or Empty Hopper	Detect blocked pellet flow or empty hopper during operation.	Alert "Motor jam" or "Hopper empty" shown. Motor stops within 2 seconds. Only one alert. Reason code logged.	Passed
TC_05	Queue and Sync Offline (No Duplicates)	Verify offline event logging and accurate cloud sync after reconnect.	Local alert created; when reconnected, only one notification is synced with the correct timestamp. No duplicates or data loss.	Passed

Table 4.1: Test case

5 BUDGET AND JUSTIFICATION

Device	Price (Rs)	Usability
Turbidity Sensor Module (SEN0189)	3800.00	Measures water clarity and contaminants
Atlas Scientific pH Sensor	7,500.00	Monitors water acidity/alkalinity
Atlas Scientific DO Sensor	6500.00	Measures dissolved oxygen levels
DHT22 Humidity Sensor	600.00	Humidity monitoring for environmental changes
ESP 32	1800.00	Controller for water quality monitoring
DS18B20	750.00	Waterproof temperature sensor
Component Total Budget	20,950.00	

Table 5.1: Budget and justification

This list covers setting up a reliable, sensor-driven maintenance assistant by the pond. The turbidity sensor (SEN0189) is a quick and cheap way to measure solids and post-feeding residue. The Atlas Scientific pH probe accurately tracks acidity and alkalinity, and the Atlas Scientific DO probe measures oxygen, which is important for fish health and a good way to start checking for aeration. The DS18B20 waterproof probe measures the temperature of the water (and compensates for pH and DO), and the DHT22 measures the humidity and temperature of the air to understand how the weather affects things. The ESP32 is a low-power controller with Wi-Fi built in. It takes samples, does basic filtering, and sends push alerts to the app. These parts, which cost Rs 20,950 together, meet the basic needs for continuous monitoring, trend-based alerts, and maintenance prompts with reliable, field-ready parts. The enclosure, cabling, and calibration buffers/solutions can be budgeted for separately as extras.

6 COMMERCIALIZATION AND MARKET FEASIBILITY

- Customers: small farms that need organized upkeep but don't want to buy a whole computerized maintenance management system
- Value: fewer missed chores, more focused help, better water quality, and better survival.
- Differentiators: trend/persistence logic → tasks, human-in-loop dosing with full audit, and integration with feeding/budget modules.

- Cost and price (approximate): core WQ node costs about Rs 20.95k (no enclosure or DO upgrades); app subscription costs per pond per month.
- Risks and ways to reduce them: sensor drift can be fixed with calibration kits and reminders; alert fatigue can be fixed with cool-downs and grouped summaries.

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8 APPENDIX

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